

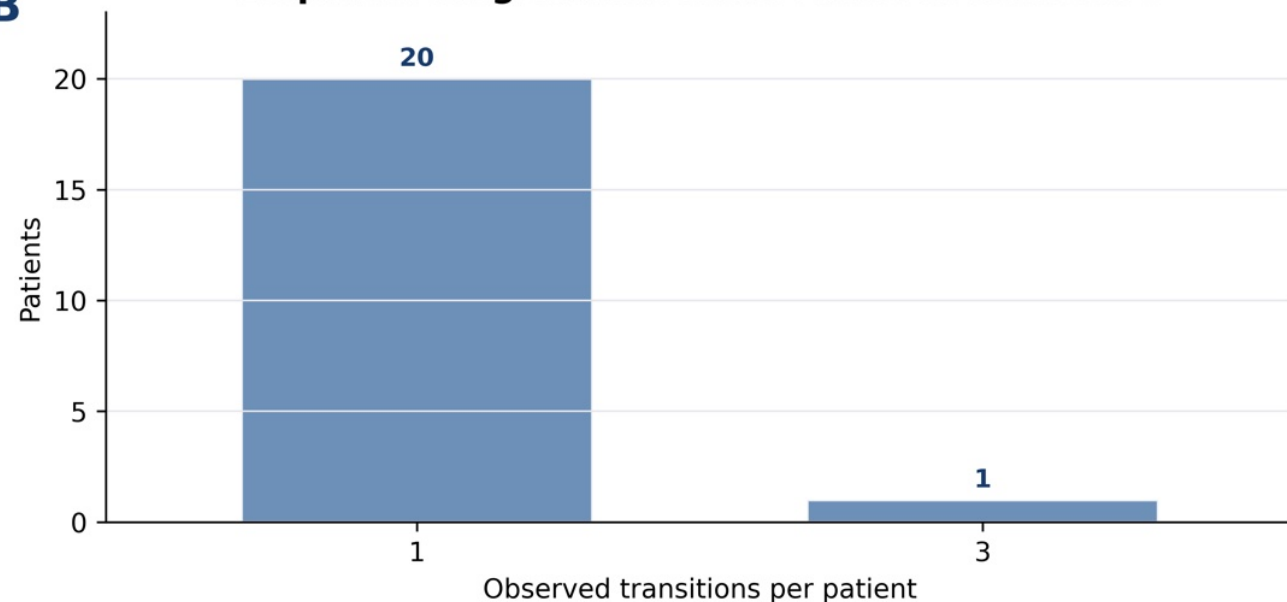
A

Empirical calibration pipeline



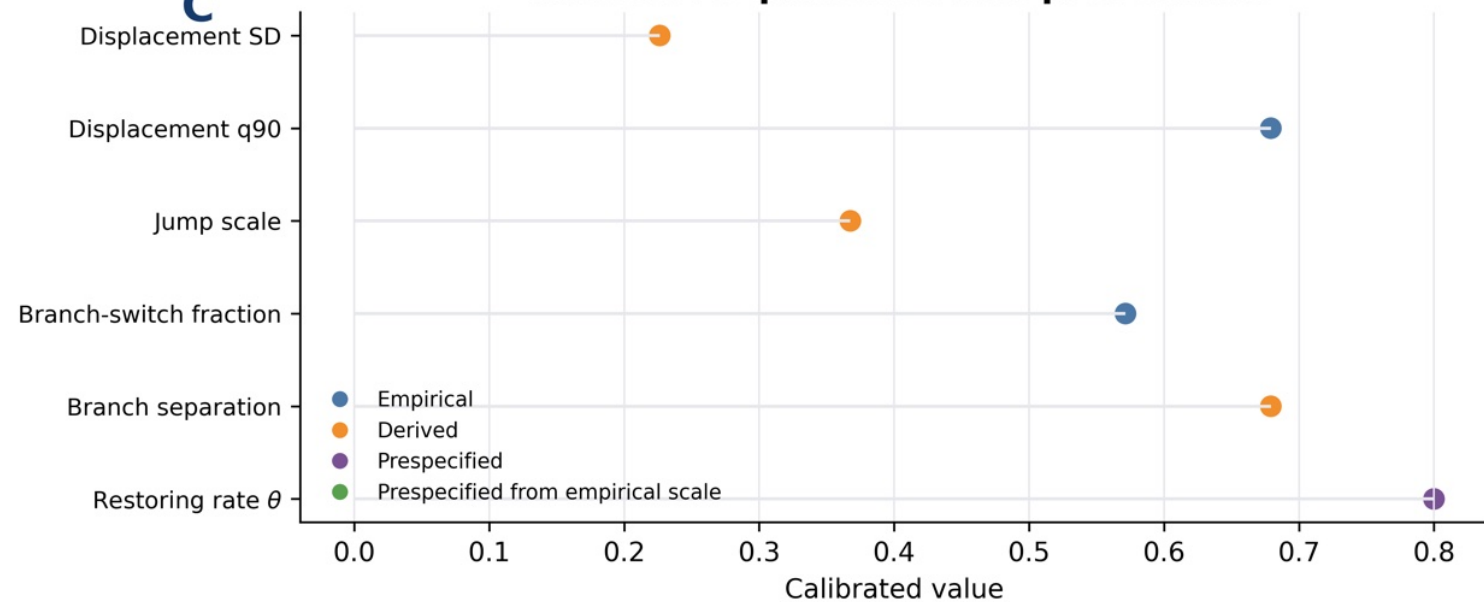
B

Empirical longitudinal observation architecture



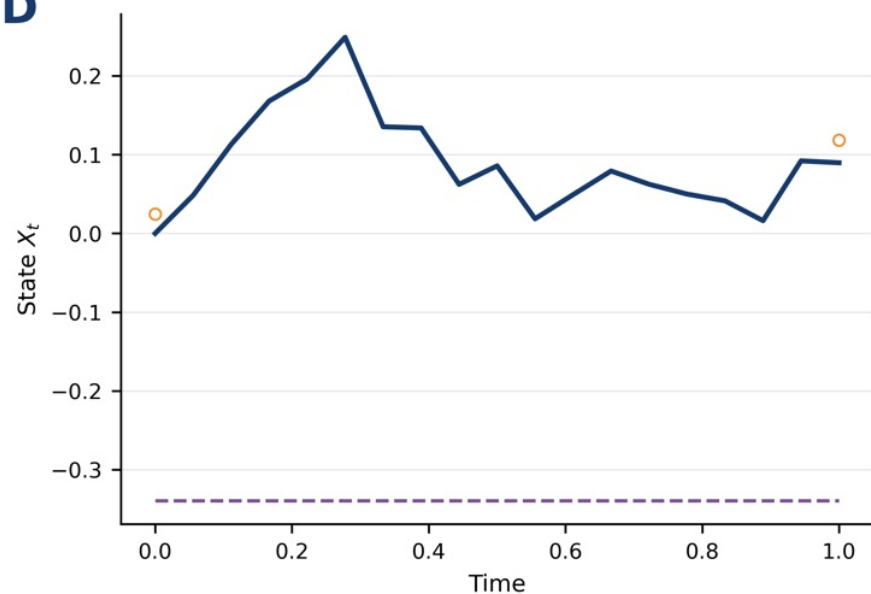
C

Calibration quantities and provenance

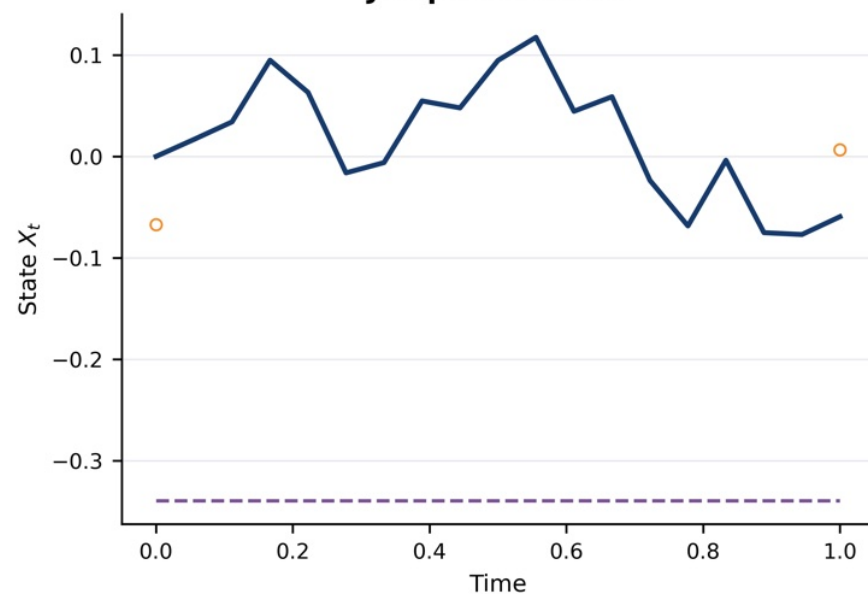


D

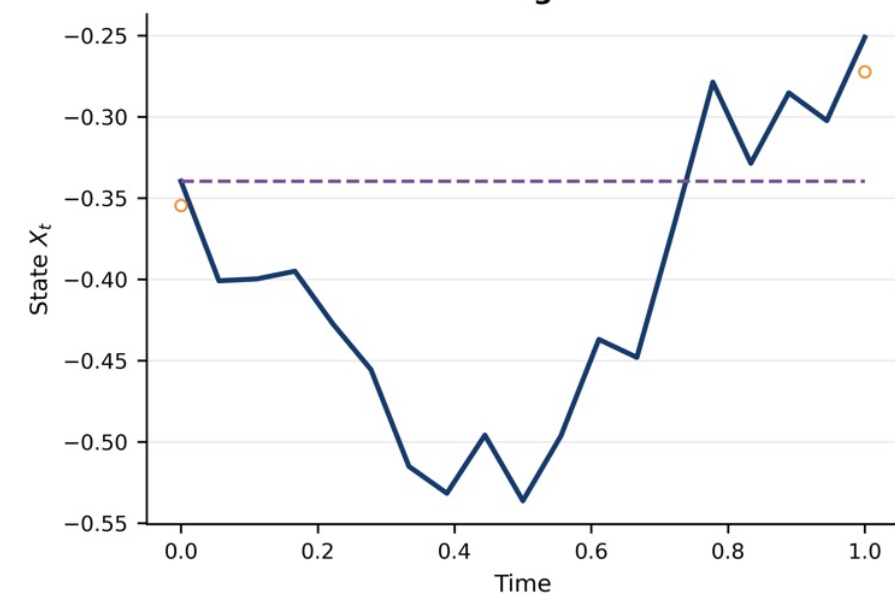
Retention-dominated



Jump-dominated

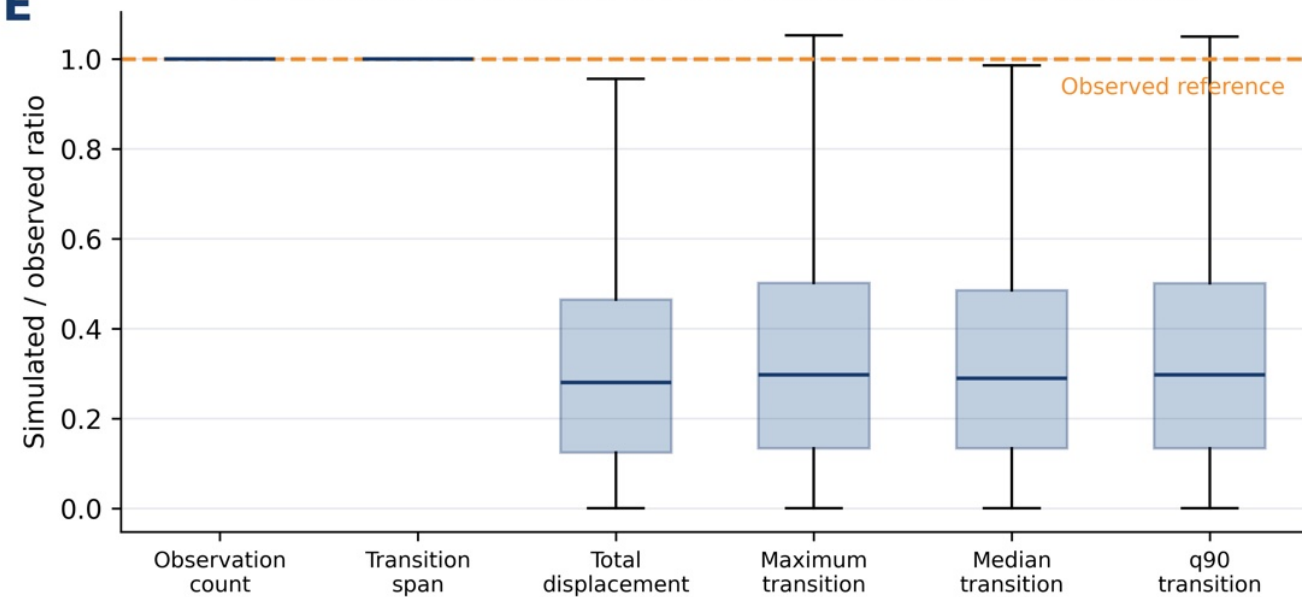


Branch reorganization



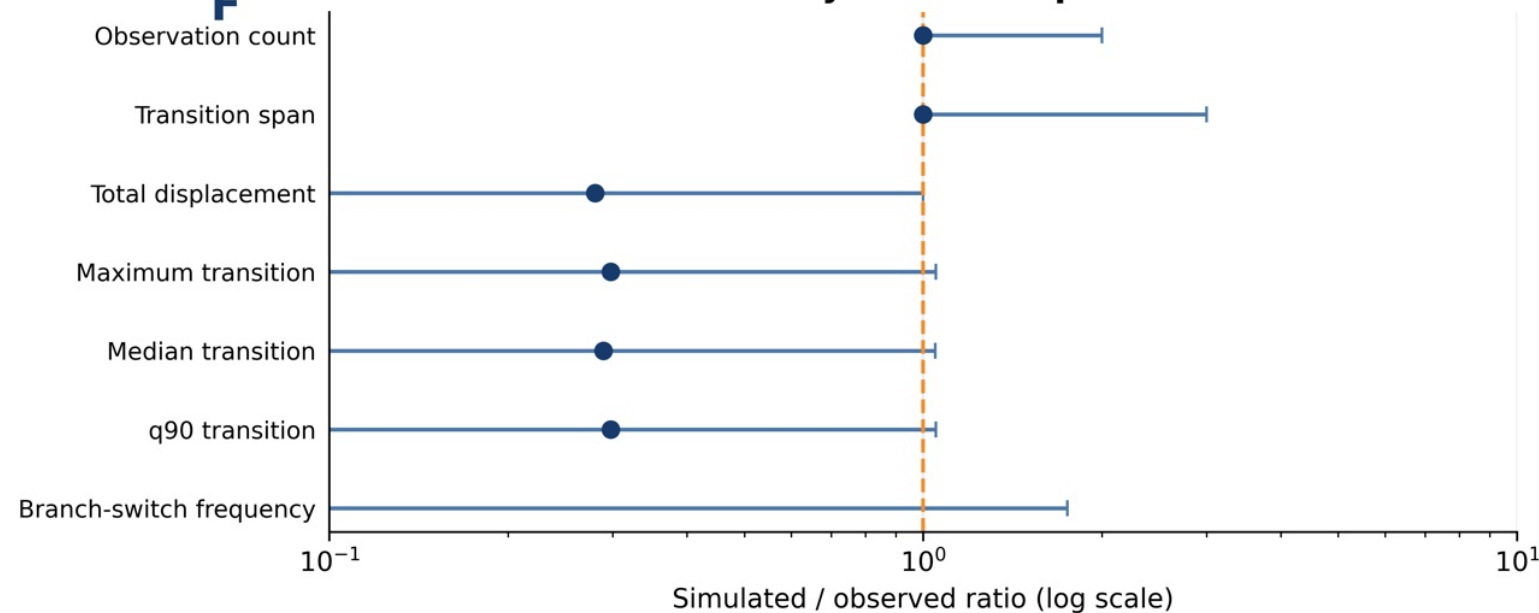
E

Observed statistics versus calibrated simulations



F

Calibration fidelity across empirical statistics



Empirical data calibrate the observation architecture and characteristic dynamical scales. Parameters not identifiable from sparse intervals are explicitly marked as prespecified.